



NOVEL PROTEASE INHIBITORS FOR THE TREATMENT OF NEUROLOGICAL DISORDERS

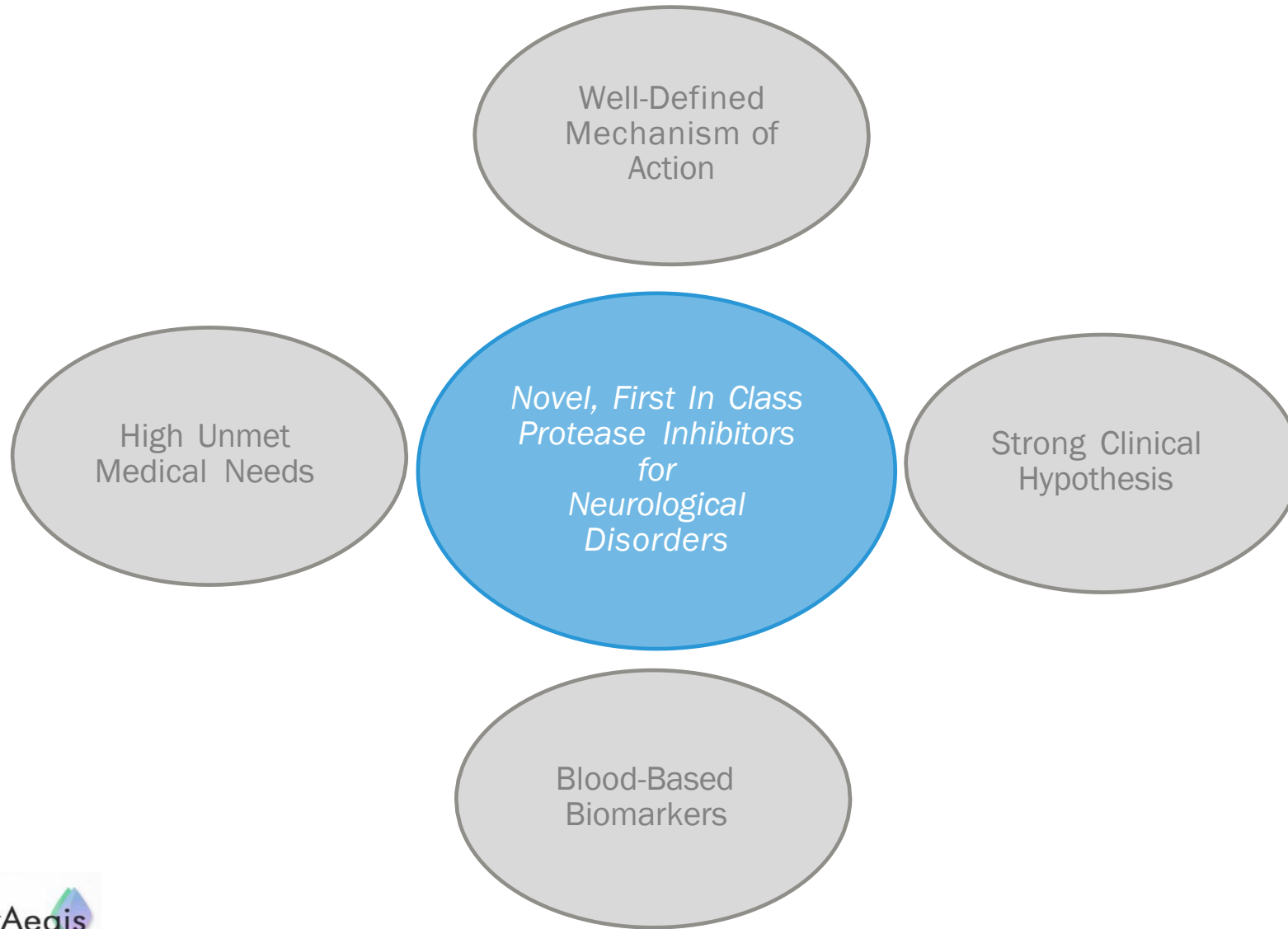
CORPORATE OVERVIEW

2024

THE NEURAEGIS OPPORTUNITY

- Portfolio of novel, first in class compounds (protease inhibitors) for treating multiple neurological disorders with significant unmet need
- Lead indications are concussion (where there is currently no FDA approved medication) and epilepsy
 - Awarded \$4.5M DOD grant for concussion therapy
 - Awarded \$700K NINDS grant for epilepsy treatment
 - Data supporting applicability in other indications, such as Loud Noise-Induced Hearing Loss and Glaucoma
- Large addressable markets
 - Over 3.8 million sports- and recreation-related concussions per year in the US (Source: www.brainline.org)
 - Epilepsy drug market to reach over \$5 billion by 2024 (Source: Grand View Research)
- Strong intellectual property
- Experienced, proven leadership team
- Significant upside potential
- Raising \$10M Series A to fund to key milestone/value inflection point of completion of Phase 1 (in concussion/TBI)

NEURAEGIS VALUE PROPOSITION



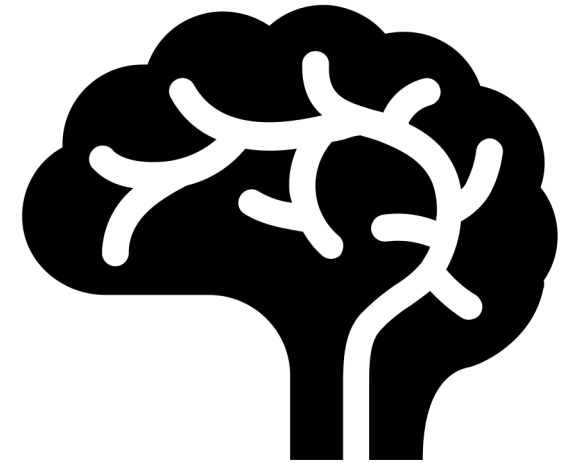
- Blockbuster potential for neurological disorders with significant unmet medical needs
- Increased likelihood of clinical success through high protease selectivity
- Potentially transformative treatment for millions of patients with neurological disorders associated with long term neurodegenerative brain effects

TARGETING NEUROLOGICAL DISORDERS WITH HUGE UNMET MEDICAL NEED

- Traumatic Brain Injury (TBI) or Concussion is a major cause of death and disability*
 - > 69,000 TBI-related deaths in the US in 2021
 - TBI may lead to long-term health problems
- TBI or concussion in childhood may affect brain development
- *Currently no approved FDA medication for TBI or concussion*
- Global epilepsy therapeutic market size was \$4.08 B in 2021 and projected to reach \$5.7 B by 2031**

*CDC

**Business Research Insights, May, 2023

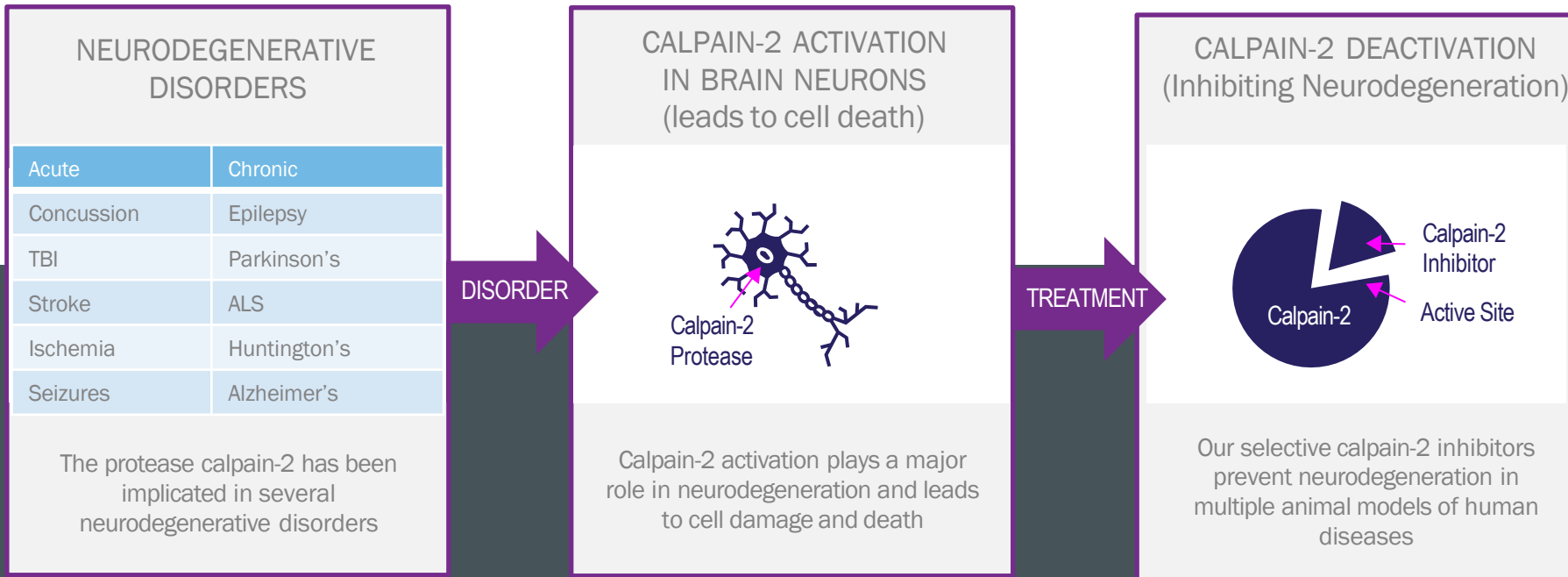


THE NEURAEGIS ADVANTAGE:

NOVEL SCIENCE ON THE VERGE OF CLINICAL PROOF OF PRINCIPLE

- Discovered the opposing roles of calpain-1 and calpain-2 in the brain, calpain-1 being neuroprotective and calpain-2 causing neurodegeneration
 - NeurAegis's lead clinical candidate NA-184 selectively inhibits calpain-2, preventing or reducing neurodegeneration
 - NA-184 is entering the clinic for TBI/concussion within 12 months
-

Preventing Neuronal Damage



INITIAL TARGET INDICATIONS

Calpain-2 has been implicated and is strongly activated in each of these disorders

Potential Indications	Current Therapies	Overall Incidence	Development Status	Current Funding
Concussion	Currently no FDA approved medications	CDC estimates 3.8 million concussions occur in the U.S. annually	• Lead candidate selected • Initiating IND-enabling program	100% funded by DOD (through IND filing)
Status Epilepticus (most extreme form of epilepsy with high mortality)	Currently no FDA approved medications	40 per 100,000 per year in the U.S. annually	• Lead optimization • Demonstrated calpain-2 inhibition significantly limits neurodegeneration after seizure activity in animal model	NINDS Award
Noise-induced hearing loss	Currently no FDA approved medications	CDC estimates 6% of adults and up to 17% of teens	• Discovery	Self Funded

INTELLECTUAL PROPERTY

- Exclusive license with Western University of Health Sciences
- Multiple pending patent applications
 - Composition of matter and therapeutic uses of calpain inhibitors
 - Novel blood biomarker for diagnosis and monitoring of acute neuronal damage
 - Potential process patents for lead compounds
- Received favorable International Search Report and Written Opinion for composition of matter claims



KEY MILESTONES

Accomplished

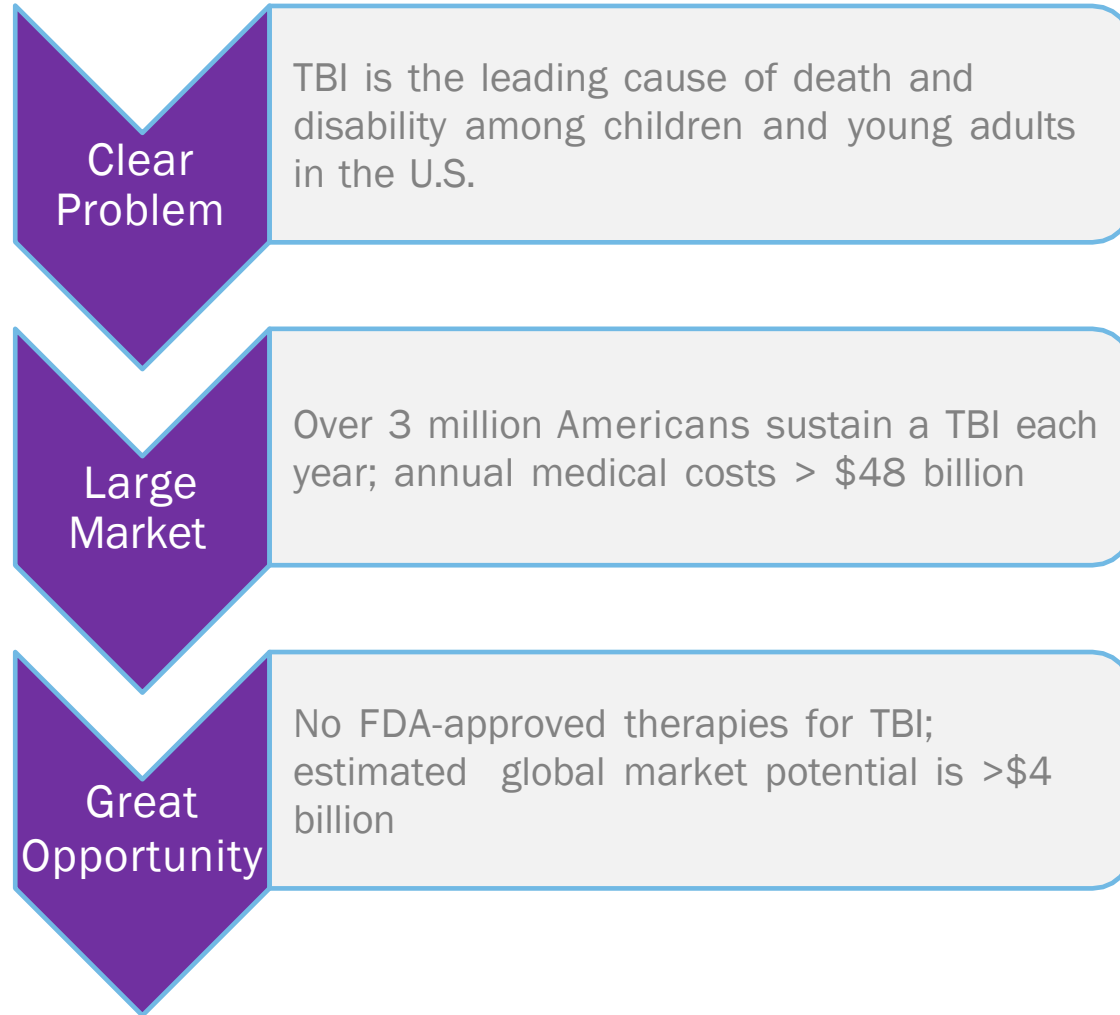
- 2016 -
 - Company formation
- 2017
 - Identified calpain-2's role in neurodegeneration
- 2018 -2019
 - Discovered novel calpain-2 protease inhibitors (NCE's)
 - Demonstrated efficacy in several preclinical animal models
 - Obtained \$3.8M DOD grant
- 2020
 - Selected clinical candidate for concussion (NA184)
 - Identified and screened novel compounds for additional indications
- 2021-2023
 - Initiate IND-enabling program for concussion/TBI

Up Next

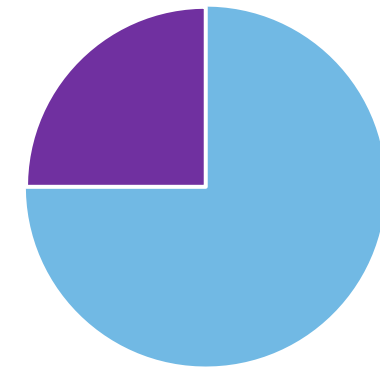
- 2024
 - Complete pre-IND studies for concussion
 - Secure funding for first-in-human clinical studies (concussion) and earlier stage programs
- 2024 - 2025
 - Select clinical candidate for status epilepticus (most severe form of epilepsy)
 - Initiate Phase 1 studies for concussion
 - Initiate IND-enabling program for status epilepticus

TBI/CONCUSSION OPPORTUNITY

UNMET MEDICAL NEED: MILD TRAUMATIC BRAIN INJURY



>3 Million Americans Sustain a TBI Annually

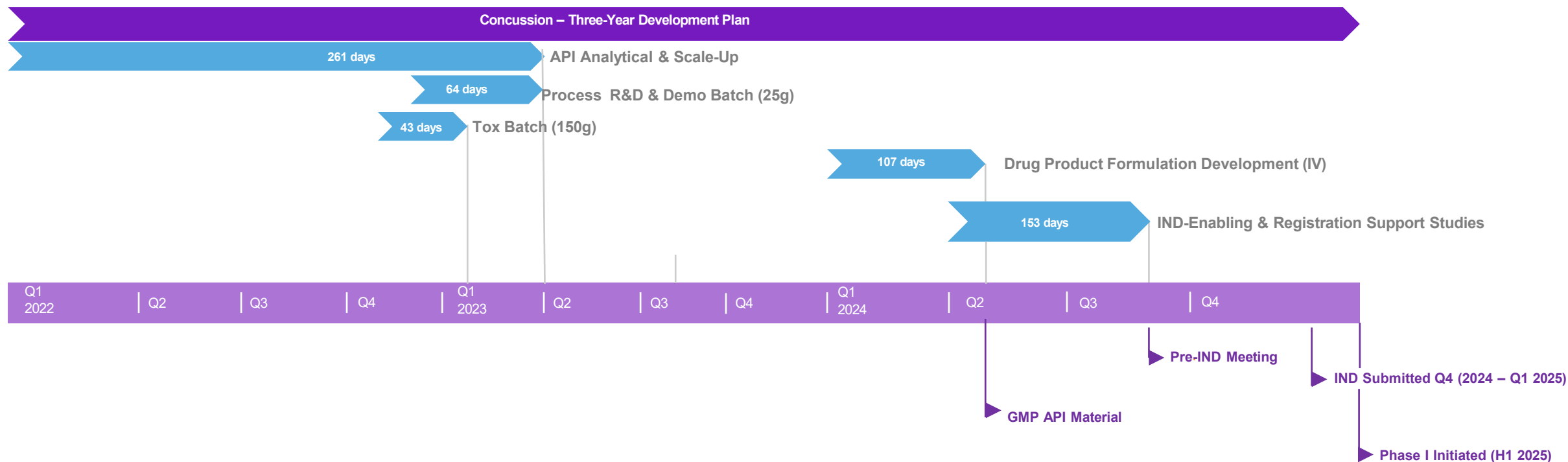


■ Concussion (75%)
■ Moderate-to-Severe TBI (25%)

OUR PATH TO PREVENTING NEURONAL DAMAGE AFTER CONCUSSION

Key Discoveries	Initial Validation of our Solution
Identified a critical <i>mechanism</i> of neuronal damage and death resulting from concussion	Demonstrated that calpain-2 activation is correlated with cell damage and death following concussion
Discovered an <i>inhibitor</i> of this neurodegenerative process	Identified novel calpain-2 inhibitors that provide significant protection when administered after concussion (animal models)
Identified blood-based <i>biomarkers</i> that directly correlates with cell death and drug target engagement	Discovered proprietary plasma biomarker (P13BP), which reflects calpain-2 activation in the brain (using human clinical samples in studies)

CONCUSSION DEVELOPMENT PROGRAM (2022-2025)



NEURAEGIS LEADERSHIP

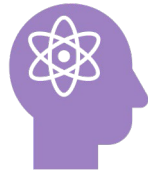
COLLECTIVELY, THE NEURAEGIS LEADERSHIP TEAM HAS...

Experience



Over 75 years of
biotechnology experience

Inventiveness



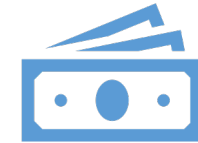
Over 400 publications, US
issued patents and pending
patents worldwide

Leadership



Led and managed over 10
biotechnology and drug
discovery companies

Market Expertise



Helped create over \$200
million in value in
associated companies

NEURAEGIS LEADERSHIP TEAM



Michel Baudry, PhD
Founder & Chief Scientific Advisor
*World-renowned neuroscientist
with over 400 publications*



Stella M. Sung, Ph.D.
President & CEO
*C-Level operating executive & VC,
helped build multiple life science firms*



Michael G. Palfreyman, PhD, DSc
Drug Development Advisor
Expert in preclinical drug development

Philippe Bey, PhD
Chemistry

Shujaath Mehdi, PhD
Biochemical Pharmacology

Kathy Fosnaugh, PhD
Program Management

Business Advisors

Lewis Geffen, Corporate Counsel; Co-Chair Life Science Practice, Mintz Levin
Peter Corless, IP Counsel; Partner, Fox Rothschild LLP

SCIENTIFIC & CLINICAL ADVISORY BOARD



Dr. Imad Najm, MD
Director of the Epilepsy Center
Cleveland Clinic



Dr. Ramon Diaz-Arrastia, MD, PhD
Director, Traumatic Brain Injury Clinical Research Center
The University of Pennsylvania
Perelman School of Medicine



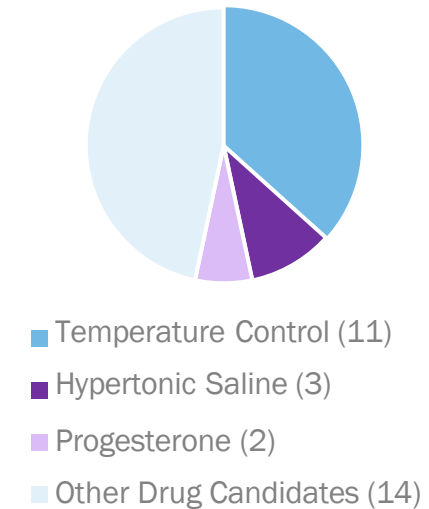
Barry Jordan, MD, MPH
Chief Medical Officer
Ranchos Los Amigos National Rehabilitation Center

NEURAEGIS COMPETITIVE ADVANTAGE

CLEAR UNDERSTANDING FOR CLINICAL TRIAL FAILURES

- All traumatic brain injury trials since 1993 have failed
- Causes of the failures:
 - Unknown mechanism for neuronal damage
 - Lack of quantitative and predictive biomarkers to guide trials
 - Subjective outcome measures
 - Patient recruitment challenges
- We are addressing each of these challenges in our clinical plan with the goal of obtaining the first FDA approved therapy for concussion

Failed TBI Clinical Trials



COMPETITIVE ADVANTAGE:

NEURAEGIS TAKES A DIFFERENTIATED APPROACH

NeurAegis's selective calpain-2 inhibitor, NAI84, is differentiable from current TBI treatments as well as other products in development

Current TBI treatment options include psychostimulants, antidepressants, antiparkinsonian agents, and anticonvulsants

TBI drugs in the development pipeline include:

- **PNT001** (Pinteon) – tau antibody
- **AST-004** (Astrocyte Pharm): adenosine receptor agonist
- **N-Acetyl-cysteine** (Neuronasal): antioxidant
- **OXE-103** (Oxeia): ghrelin
- **VAS203** (Vasopharm) - inhibitor of nitric oxide synthase
- **NeuroSTAT** (NeuroVive Pharmaceutical) - proprietary formulation of ciclosporin; targeting mitochondria
- **CEVA101** (Cellvaton) - Autologous Bone Marrow-derived Mononuclear Cells (BMMNCs)

Favorable Market Dynamics

- TBI drugs in pipeline with novel MoA are still years from commercialization—expected in the 2024-2026 timeframe
- Blockbuster potential for first in class or best in class therapeutics, and room for multiple “winners”
- VAS203 expected to reach peak revenues of \$2.5B by 2030
- Potential for relatively short development and regulatory timelines
- In 2019, NeuroSTAT received Fast Track designation from the FDA

SERIES A FINANCING

SUMMARY OF PROPOSED TERMS FOR SERIES A RAISE

- \$10 Series A Preferred financing
 - Funds to key milestones/value inflection points (Ph 1 in concussion)
 - 2 years operating runway
- Fully diluted pre-money valuation= \$15M (\$1.64/share)
- Liquidation preference
 - First pay one times the Original Purchase Price plus declared and unpaid dividends on each share of Series A Preferred; (or, if greater, the amount that the Seed Preferred would receive on an as-converted basis). The balance of any proceeds shall be distributed pro rata to holders of Common Stock.
- Weighted average anti-dilution protection
- One Board or Observer seat (if desired) for Series A Preferred investor class

COMPARABLES

Amylyx Pharmaceuticals, Inc.

Company focused on the discovery, development and commercialization of innovative new treatments for the neurodegenerative community. Scientific focus is to potentially reduce neuronal cell death.

Public company (NYSE: ANVS) Market cap at high >\$2.6B

Annovis Bio

Clinical stage, drug platform company focused primarily on drugs for chronic neurodegeneration—AD, its orphan indication AD-DS and PD. Additionally, Annovis has a compound to treat acute neurodegeneration—traumatic brain injury and stroke—and a third compound for advanced AD.

Public company (NYSE: ANVS) Market cap at high >\$1B

Remedy Pharmaceuticals

Intravenous formulation for acute central nervous system conditions such as stroke (ischemic and hemorrhagic), traumatic brain injury.

Acquired By: Biogen May 15, 2017 \$120M

Plus, several private companies (e.g. Oxeia Biopharmaceuticals) and psychedelics companies (hot sector)

SUMMARY OF INVESTMENT OPPORTUNITY

- Novel, first in class protease inhibitors have blockbuster potential
 - Addressing multiple neurological disorders
 - No current FDA-approved medication—huge unmet medical need!
- Strong Intellectual Property
- Experienced Leadership Team
- Significant Upside Potential
 - Key milestones achieved on grant funding (validative and non dilutive)
 - First institutional round
 - Exit comparables suggest 10x return potential



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THANK YOU!